Gaussian Process Classification: How kernels shape decision boundaries and uncertainty

A Tutorial Report

Tutorial focus
How kernel choice changes the decision boundary, smoothness and uncertainty of a Gaussian Process Classifier.
Submission contents: runnable notebook, PDF tutorial report, requirements file, README and license attached in the github repository: https://github.com/Avinash24176747/Individual-coursework-Machine-Learning-Tutorial .

Abstract

Gaussian process classification, or GPC, is a probabilistic binary classification procedure. It assigns class names, but it assigns probabilities. This is one of the reasons why it is useful. One of the narrow sections of GPC is examined in this report. It questions the alteration of the decision boundary and the model's uncertainty due to the kernel choice. That narrows down the subject, making it practical as well. They are a noisy pair of moons: data and the Breast Cancer Wisconsin data. The kernels compared are RBF, Matern and linear. As the report shows, kernels are not a minor detail that can be overlooked. They determine the type of functions that the model believes can be reasonable prior to the data being observed [1][3]. In straightforward terms, the kernel will modify how smooth, local or straight the end result will be.

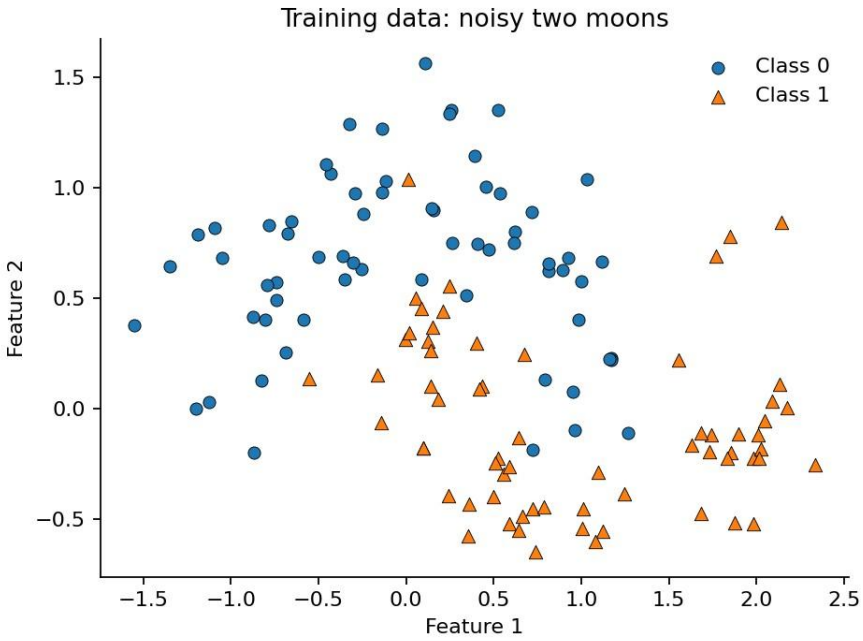

Figure 1. Training split of the noisy two-moons dataset used to make kernel behaviour visually obvious. Marker shape distinguishes classes as well as colour.

1. Technique and focus

The method used in this tutorial is Gaussian Process Classification. A Gaussian process may be regarded as a distribution of the functions which may exist, not only a single line or curve [1]. A latent function is learned in the first stage of classification. Then, the latent value is converted into a

probability distribution over classes [2]. This is implemented in scikit-learn using a Laplace approximation, enabling the method to be applied in practice without full inference [3].

This topic is quite wide. As a result, the tutorial does not attempt to explain all the details of Gaussian process theory. The focus stays on kernels. It is the most helpful option in teaching. Many of the modelling assumptions are sunk in the kernel. It informs the model about which shapes are plausible. It may be a very smooth shape that is desired. Or a rougher one. Or almost a straight split. Understanding the kernel helps students understand GPC's behaviour.

2. Why Gaussian Process Classification is worth teaching

GPC is a powerful subject of the tutorial for three reasons. First, it reveals a crucial difference with most introductory classifiers: it yields probabilities and uncertainty estimates rather than only hard labels [1][3]. Second, it provides a pathway directly to kernel thinking, also found in support vector machines and other non-parametric techniques. Third, it favours pictorial description. The model behaviour is visible in decision boundaries, probability fields, and uncertainty maps, which are extremely efficient for students [4].

The method is also at a good level of difficulty in this determination. It is more advanced than a conventional logistic regression or a basic multilayer perceptron; however, it can be described thoroughly and implemented in a small notebook. Such a balance makes the tutorial feel deep without being overly mathematically heavy, so the quality of the teaching is not compromised.

3. Demonstration design

To illustrate the kernel's impact, the notebook begins with a rather noisy two-moon dataset. The classes are bound around, and a linear boundary can be seen. I compare three of them, i.e., RBF, Matern 1.5 and 1.5, as well as the linear kernel. The original set of experiments does not involve a lengthy optimisation step; instead, it uses fixed kernel settings to enable quick, interpretable comparisons. This means the figures reveal the kernel's assumptions.

In the display outputs, there are two interconnected images. The former depicts the probability surface P(class 1). The second illustrates an uncertainty map defined as 1 - max_p (c|x). With binary classification, the greatest value is at the point where the model is most sought after, near a 50/50 decision. Such a property can be successfully used pedagogically: learners get to visualise the taxonomies and regions of model hesitation.

After comparing with the kernel, the notebook conducts a deep but narrow exploration of the RBF length scale. This is one of the key lessons of kernel methods: when length scales are small, the classifier is sensitive to local detail, but when they are large, the decision function varies more gradually [1]. Finally, the notebook uses breast cancer data from Wisconsin to demonstrate that the concepts can be generalised to a binary classification problem.

Table 1. Two-moons test performance from the notebook. Lower Brier score and lower log loss indicate better probability quality.

Kernel	Accuracy	Brier	Log loss
RBF	0.943	0.081	0.307
Matérn ν=1.5	0.957	0.074	0.295

Linear	0.886	0.081	0.28

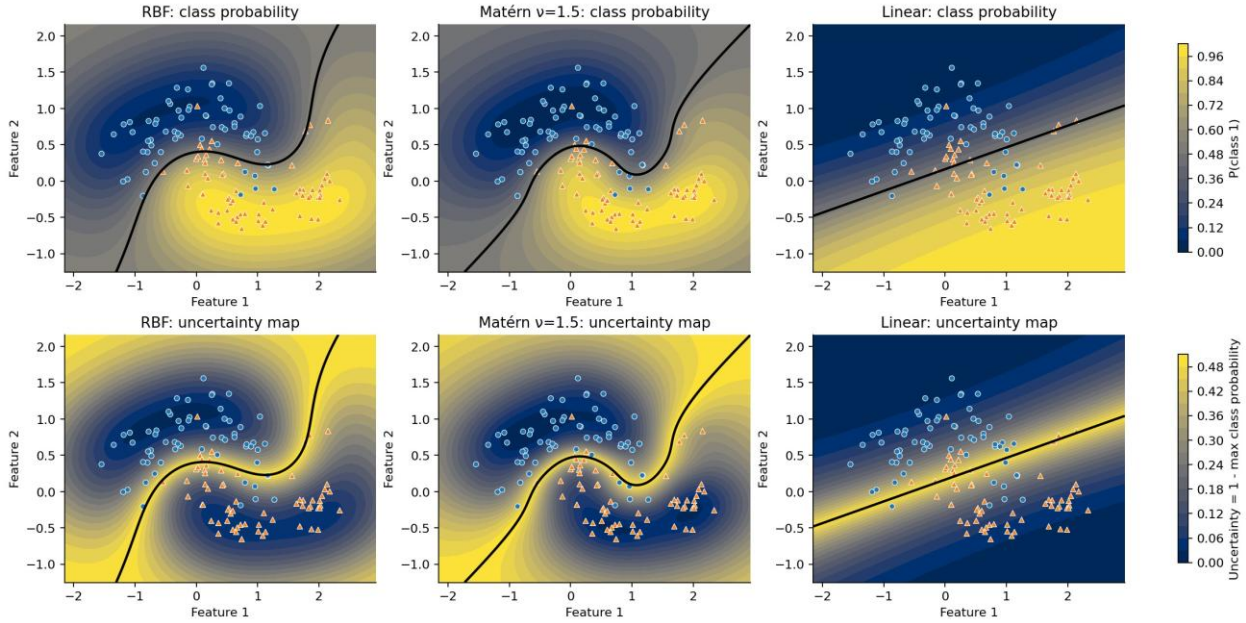

Figure 2. Kernel comparison on the same training set. The top row shows class probability surfaces and the bottom row shows uncertainty maps. RBF and Matérn can bend around the moons; the linear kernel cannot.

4. What the results show

The experiment with two moons demonstrates that there are recognisably distinct geometries due to different kernels. The RBF kernel produces a smooth, curved edge, which is very good at tracking the crescent shape. The Matern kernel similarly bends around the classes but with a slightly coarser local variation. The linear kernel, in turn, is unable to wind around the moons and thus leaves an unexplained visible structure. It is precisely this type of analogy that makes the concept of a kernel teachable.

The deeper point is that every kernel reflects a previous belief about possible functions. The kernel used in the RBF is highly smooth. The Matern family eases that supposition and allows more rough behaviour [1]. A linear kernel assumes that a linear split suffices. These assumptions can be read directly in the notebook, not just by reading the equations.

The same message is supported by the length-scale experiment. The length scale is small; thus, the boundary is more localised and more flexible. The boundary is curved, more stable, and has a minimal length scale. A very large length scale leads to a too-rigid classifier that begins to smooth over useful structure.

Notably, the most balanced probabilistic behaviour does not necessarily coincide with the maximum accuracy. This is the reason why the Brier length scale score and log loss are also reported alongside accuracy by the notebook.

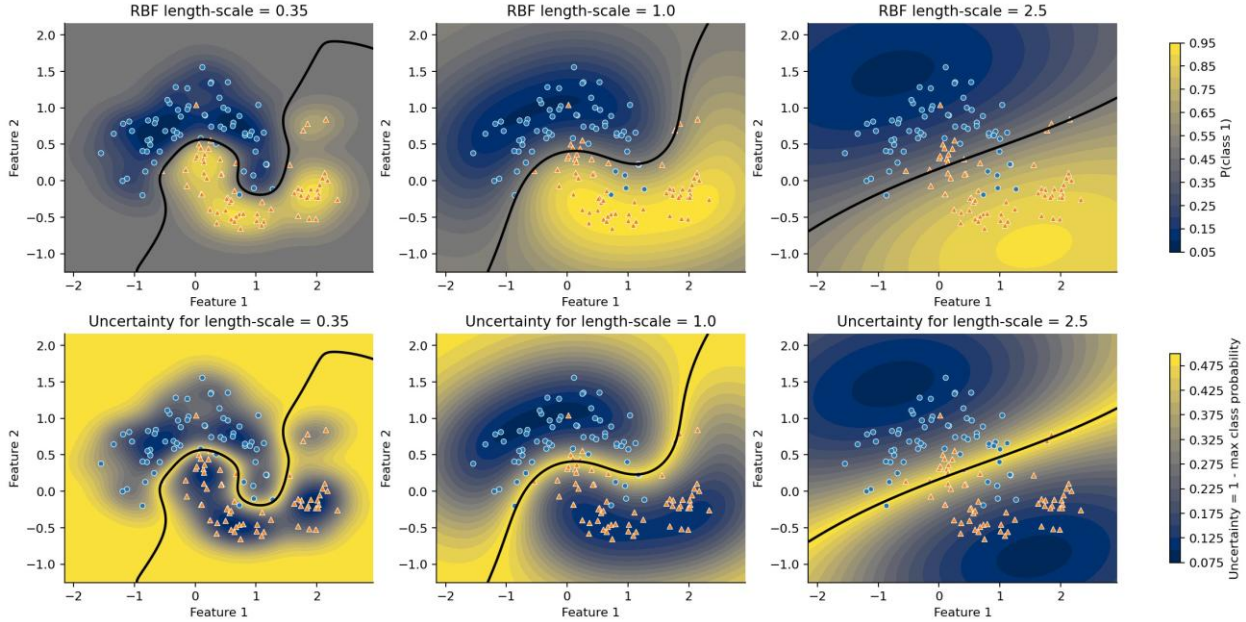

Figure 3. RBF length-scale comparison. Smaller length-scales create more local, flexible boundaries; larger length-scales force smoother, more global behaviour.

5. Real-data example and practical lessons

A more sophisticated lesson may be included with the breast-cancer experiment, though: a more complex kernel is not necessarily better. In this dataset, the linear kernel performs well and is superior to certain non-linear models in terms of accuracy and probability. That is important to teach since it avoids one of the popular misconceptions that sophisticated techniques must always follow the most flexible model. In some cases, the data geometry is sufficiently close to linear after standardisation, and a simpler kernel is preferable.

The effective message is thus moderate. GPC is attractive when the dataset is small to medium, when it is important to make probability estimates, and when it is valuable to make decisions that are sensitive to uncertainty. When the dataset is large, it is less attractive, since methods based on Gaussian processes become computationally costly as the sample size increases [1][3]. The tutorial presents both sides of the case to help students understand when to use the method, rather than merely learning how to execute the code.

Table 2. Breast-cancer test performance from the real-data section. Here, the simpler linear kernel performs best, showing that more flexible kernels are not automatically superior.

Kernel	Accuracy	Brier	Log loss
Linear	0.988	0.02	0.079
RBF ■=2.0	0.965	0.085	0.31
RBF ■=1.0	0.971	0.204	0.597
Matérn ■=1.0	0.959	0.176	0.539

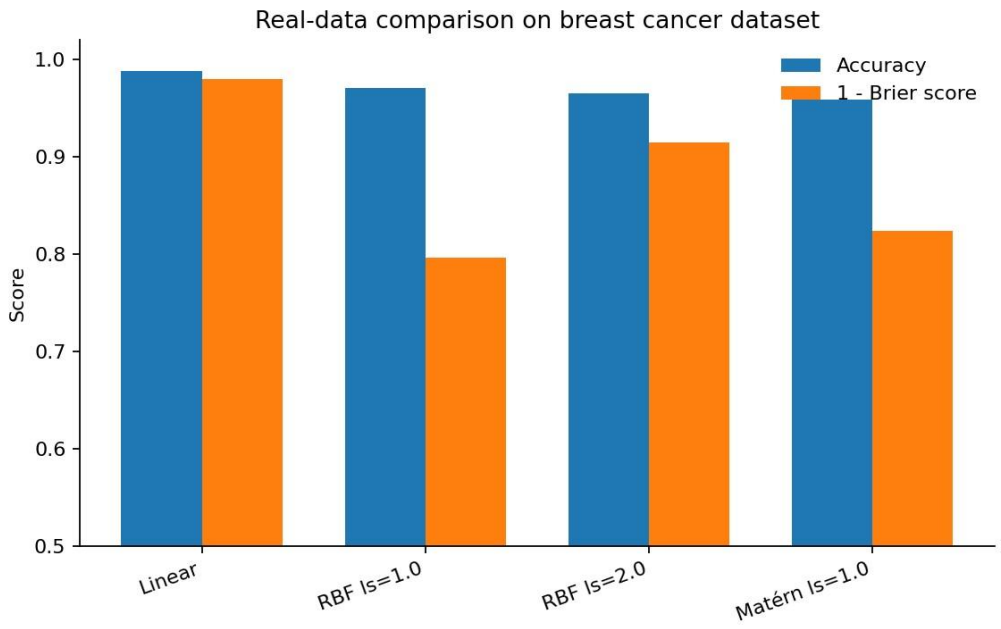

Figure 4. Real-data metric comparison. Accuracy and probability quality should be read together when the model will support risk-aware decisions.

6. Code quality, creativity and accessibility

The code was developed on the basis that it can be executed by another user without much inconvenience. It uses standard Python packages, mostly NumPy, pandas, matplotlib, and scikit-learn. The datasets are also available in scikit-learn, so they do not require external download. It contains a requirements file. It also includes a short READM. This was supplemented by an MIT licence to make reuse more apparent.

Certain accessibility measures were also implemented. The plots are displayed using the cividis colour map, which is easier for many colour-blind readers than some default colour maps. The classes are marked with different marker shapes; thus, colour alone does not carry all the work. The report and notebook provide what is expected to be observed in each figure. That assists the readers who might not be able to get everything using the image. There is still more that can be added, such as alt text for all exported images, but as it stands, the current version actually makes an attempt.

7. Conclusion

Gaussian Process Classification is a good method to learn because it interrelates predictive probabilistic forecast, kernel design, and visual analysis within a single workflow. This tutorial narrows a broad subject into an understandable, applicable lesson by addressing kernel selection options. The code and figures demonstrate that kernels do not just tune performance; they specify the types of boundaries the classifier is confident are practical. The main lesson is that understanding the impact of kernel choice leads to deeper insight, making graduates more capable of using GPC and kernel methods effectively in their own programmes.

References

[1] C. E. Rasmussen and C. K. I. Williams, Gaussian Processes for Machine Learning, MIT Press, 2006.

[2] C. K. I. Williams and D. Barber, “Bayesian Classification with Gaussian Processes,” IEEE Transactions onPattern Analysis and Machine Intelligence, vol. 20, no. 12, pp. 1342–1351, 1998. doi: 10.1109/34.735807.

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[4] J. Görtler, R. Kehlbeck and O. Deussen, “A Visual Exploration of Gaussian Processes,” Distill, 2019.

[5] scikit-learn developers, “Breast Cancer Wisconsin dataset,” scikit-learn documentation.